

☒ Easy (10 Questions)

1.

Which TCP flag initiates a connection?

- A) ACK
- B) SYN
- C) FIN
- D) RST

Answer: B

Explanation: SYN flag is used to start the TCP three-way handshake.

2.

What is banner grabbing used for?

- A) Blocking network traffic
- B) Gathering service information from a server
- C) Encrypting data
- D) Crashing servers

Answer: B

Explanation: Banner grabbing collects information like service names and versions.

3.

Which technique hides an attacker's real IP address?

- A) Banner grabbing
- B) Proxy servers
- C) Port scanning
- D) Packet sniffing

Answer: B

Explanation: Proxy servers relay traffic and mask attacker IPs.

4.

HTTP tunneling allows attackers to:

- A) Send HTTP requests only
- B) Encapsulate other protocols inside HTTP to bypass firewalls
- C) Encrypt emails
- D) Monitor wireless signals

Answer: B

Explanation: HTTP tunneling encapsulates traffic to evade firewall restrictions.

5.

IP spoofing involves:

- A) Changing destination IP
- B) Changing source IP in packets
- C) Encrypting IP headers
- D) Blocking IP addresses

Answer: B

Explanation: IP spoofing forges the source IP to impersonate others.

6.

Enumeration gathers information about:

- A) Network devices and user accounts
- B) Only network topology
- C) Encrypted passwords
- D) Firewall rules

Answer: A

Explanation: Enumeration extracts detailed system info like users and services.

7.

Which password cracking method tries all possible combinations?

- A) Dictionary attack
- B) Brute force attack
- C) Rainbow table attack
- D) Social engineering

Answer: B

Explanation: Brute force exhaustively tries every combination.

8.

Windows password hashes are stored in:

- A) Registry
- B) SAM file
- C) Active Directory
- D) Hosts file

Answer: B

Explanation: The SAM file stores Windows password hashes.

9.

SMB relay attack exploits:

- A) FTP protocol
- B) SMB authentication messages
- C) DNS servers
- D) HTTP headers

Answer: B

Explanation: SMB relay attacks intercept and forward SMB authentication.

10.

A NetBIOS DoS attack targets:

- A) Web servers
- B) Windows networking services
- C) Email servers
- D) DNS servers

Answer: B

Explanation: NetBIOS DoS floods Windows network services causing denial of service.

☒ Medium (20 Questions)

11.

Which TCP flag is used to immediately terminate a connection?

- A) FIN
- B) SYN
- C) RST
- D) ACK

Answer: C

Explanation: RST flag forces an abrupt connection reset.

12.

Active OS fingerprinting involves:

- A) Listening to traffic only
- B) Sending crafted packets and analyzing responses
- C) Crashing the target system
- D) Sending spam emails

Answer: B

Explanation: Active fingerprinting interacts directly with the target.

13.

How do proxy servers help attackers?

- A) Increase bandwidth
- B) Hide the attacker's IP and anonymize requests
- C) Encrypt all traffic
- D) Prevent malware

Answer: B

Explanation: Proxies mask the source IP making tracing difficult.

14.

Which of the following is a sign of HTTP tunneling?

- A) Traffic on port 22
- B) Non-HTTP protocol data inside HTTP packets
- C) ICMP echo requests
- D) FTP commands inside HTTP

Answer: B

Explanation: Encapsulation of other protocols inside HTTP is HTTP tunneling.

15.

Blind IP spoofing means:

- A) Sending packets without expecting replies
- B) Intercepting replies first
- C) Using the target's IP as source
- D) Spoofing DNS records

Answer: A

Explanation: Blind spoofing occurs without seeing responses from the target.

16.

Which tool can be used for enumeration?

- A) Nmap
- B) Netcat
- C) Enum4linux
- D) John the Ripper

Answer: C

Explanation: Enum4linux gathers SMB and Windows info for enumeration.

17.

A dictionary attack:

- A) Tries every possible password
- B) Uses a list of common passwords
- C) Uses rainbow tables
- D) Uses brute force

Answer: B

Explanation: Dictionary attack uses precompiled lists of likely passwords.

18.

Which attack captures SMB credentials by redirecting logon requests?

- A) SMB relay
- B) Man-in-the-middle DNS attack
- C) ARP poisoning
- D) SQL injection

Answer: A

Explanation: SMB relay intercepts and forwards authentication to capture credentials.

19.

Countermeasure against SMB relay attack includes:

- A) Disable SMB signing
- B) Use NTLMv1
- C) Enable SMB signing and NTLMv2
- D) Use default passwords

Answer: C

Explanation: SMB signing and NTLMv2 mitigate relay attacks.

20.

Which TCP flag combination is used in an XMAS scan?

- A) SYN only
- B) FIN, URG, PSH
- C) RST only
- D) ACK only

Answer: B

Explanation: XMAS scan sets FIN, URG, and PSH flags.

21.

IP spoofing can be detected by:

- A) Checking for inconsistent packet source IPs and sequence numbers

- B) Only using firewalls
- C) Disabling ports
- D) Using antivirus

Answer: A

Explanation: Anomalies in packet headers can indicate spoofing.

22.

SMB stands for:

- A) Simple Mail Block
- B) Server Message Block
- C) Secure Mail Broker
- D) System Management Bus

Answer: B

Explanation: SMB is used for file and resource sharing on Windows.

23.

Which attack floods a network with excessive traffic from multiple sources?

- A) Phishing
- B) DDoS
- C) XSS
- D) SQL Injection

Answer: B

Explanation: Distributed DoS attacks overwhelm resources with traffic.

24.

Password hashes stored in Windows using LM Hash are vulnerable because:

- A) They are salted
- B) They are weak and easily cracked
- C) They use SHA-256
- D) They are encrypted with AES

Answer: B

Explanation: LM Hash is weak and considered insecure.

25.

Traceroute uses:

- A) ICMP or UDP packets
- B) TCP SYN packets only
- C) HTTP requests
- D) DNS queries

Answer: A

Explanation: Traceroute commonly sends ICMP or UDP packets to map paths.

26.

What does the PSH TCP flag indicate?

- A) Close connection
- B) Immediate data delivery to application
- C) Acknowledgment of data
- D) Synchronize connection

Answer: B

Explanation: PSH flag pushes data to the receiving application immediately.

27.

Which attack can occur if an attacker uses IP spoofing in a DoS?

- A) Reflection attack
- B) Phishing
- C) Cross-site scripting
- D) SQL Injection

Answer: A

Explanation: Reflection attacks use spoofed IPs to amplify DoS.

28.

Which protocol is vulnerable to SMB relay attacks?

- A) FTP
- B) SMB
- C) DNS
- D) HTTP

Answer: B

Explanation: SMB relay exploits the SMB protocol's authentication.

29.

NetBIOS is primarily used for:

- A) File and printer sharing on Windows networks
- B) Web browsing
- C) Email transfer
- D) Secure tunneling

Answer: A

Explanation: NetBIOS supports Windows networking services.

30.

Which countermeasure reduces the impact of NetBIOS DoS attacks?

- A) Disable NetBIOS if not needed
- B) Use default passwords
- C) Enable open shares
- D) Disable firewalls

Answer: A

Explanation: Disabling unused NetBIOS reduces attack surface.

☒ Hard (10 Questions)

31.

In TCP, what is the purpose of the URG flag?

- A) Marks urgent data that should be processed immediately

- B) Terminates the connection
- C) Starts the handshake
- D) Acknowledges data

Answer: A

Explanation: URG flag indicates urgent pointer data for immediate processing.

32.

How does passive OS fingerprinting differ from active?

- A) It sends crafted packets
- B) It only observes traffic without sending probes
- C) It crashes the target
- D) It modifies packet headers

Answer: B

Explanation: Passive fingerprinting listens to traffic to identify OS without interaction.

33.

What technique do attackers use to chain proxies?

- A) VPN tunneling
- B) Proxy chaining to increase anonymity by routing through multiple proxies
- C) DNS poisoning
- D) ARP spoofing

Answer: B

Explanation: Proxy chaining routes traffic through several proxies for anonymity.

34.

HTTP tunneling can be detected by:

- A) Analyzing packet payloads for encapsulated non-HTTP protocols
- B) Blocking port 80
- C) Using DNS logs only
- D) Encrypting HTTP traffic

Answer: A

Explanation: Deep packet inspection can reveal non-HTTP data inside HTTP packets.

35.

Which Windows hash type was deprecated due to insecurity?

- A) NTLMv2
- B) LM hash
- C) SHA-1
- D) bcrypt

Answer: B

Explanation: LM hash is insecure and deprecated.

36.

SMB relay attacks exploit which vulnerability?

- A) Lack of SMB signing and weak authentication protocols
- B) Weak passwords
- C) Open ports
- D) DNS spoofing

Answer: A

Explanation: Without SMB signing, authentication messages can be relayed.

37.

A common tool used for SMB relay attacks is:

- A) Responder
- B) Nmap
- C) Wireshark
- D) Metasploit Framework

Answer: A

Explanation: Responder is widely used to perform SMB relay and poisoning.

38.

Which TCP scan sends RST after SYN to avoid full connection?

- A) SYN scan
- B) Xmas scan
- C) FIN scan
- D) ACK scan

Answer: A

Explanation: SYN scan sends SYN and then RST to avoid completing handshake.

39.

What does a DDoS protocol attack target?

- A) Protocol weaknesses like SYN flood or UDP flood
- B) Application layer only
- C) Physical layer
- D) Email servers

Answer: A

Explanation: Protocol attacks exploit weaknesses in TCP/IP stack.

40.

Password cracking with rainbow tables requires:

- A) Precomputed hash tables to reverse hashes quickly
- B) Brute force attempts
- C) Guessing passwords
- D) Manual entry

Answer: A

Explanation: Rainbow tables are precomputed hashes for rapid password recovery.